OMNI-FUSION — Video Presentation Script

Byte2Beat Kaggle Challenge • Hack4Health • Cardiovascular AI Track

Duration: Under 3 Minutes (~2 min 50 sec) | Format: Screen-Recording with Voice-Over | July 2026

🎤 **Presenter Tip:** Speak at a calm, confident pace (~155 wpm). Timestamps on the left are targets. Blue cue boxes tell you what to show on screen. Italicised text is what you say.

0:00 – 0:18 1. Hook & Problem Statement

🎬 *SHOW: Title card "Omni-Fusion: Multimodal Cardiovascular AI". Then transition to landing page at localhost:3000.*

*"Cardiovascular disease kills over **17.9 million people every year**. In overcrowded emergency rooms, doctors must make triage decisions with **fragmented patient data** — ECG readings in one system, lab reports in another, and past medical history often completely missing. **Omni-Fusion** is a multimodal AI platform that solves this by fusing all three data streams into a single, explainable cardiovascular risk score."*

0:18 – 0:42 2. Architecture & How It Works

🎬 *SHOW: The three feature cards on the landing page (Multimodal AI, Enterprise Security, Clinical Workflow). Then briefly show the architecture diagram from README or the Kaggle manuscript.*

*"Under the hood, Omni-Fusion runs three specialised neural networks in parallel. A **1D ResNet** processes raw 12-lead ECG waveforms at 500 Hz — trained on the public **PTB-XL** dataset. A **deep MLP** processes 11 clinical vitals and lab values like Troponin, Creatinine, and SpO2. And a **GRU recurrent network** captures longitudinal patient history from past hospital visits — trained on **MIMIC-IV** clinical data. All three embeddings are concatenated and fed into a fusion classifier. If a patient's historical records are missing, our **Dynamic Routing Masking** automatically zeros out that embedding — no crash, no separate model needed. The system just works. We also built an **Equity-Aware Loss** function that penalises the model when accuracy diverges between male and female patients, closing the gap from **12.5% to under 1.5%**."*

0:42 – 1:05 3. Patient Sign-In, Onboarding & Data Upload

🎬 *SCREEN RECORD: Click "Demo Accounts" on login page → click a demo patient (e.g. Ramesh Iyer — Critical) → land on Patient Dashboard → show risk score card, trend chart, and ABDM sandbox card.*

*"Let me show you the platform live. On the login page, we have **one-click demo accounts** — five patients and three doctors pre-seeded with realistic backdated histories. I'll log in as Ramesh, a critical-risk patient. The **Patient Dashboard** greets me with my latest risk score, risk category, total assessments count, and a **longitudinal risk trend chart** plotted over time. There is also an **ABDM sandbox card** showing a simulated Ayushman Bharat ABHA-ID — we've built the consent flow and FHIR R4 mapping for India's national health infrastructure. Now let me start a new assessment."*

`1:05 – 1:40` 4. Running Inference & Explainability

> 🎬 *SCREEN RECORD: Navigate to /patient/assessment/new → Upload a blood report PDF or CSV (Step 1) → Optionally upload an ECG image (Step 2) → Click "Run Multimodal Inference" (Step 3) → Show the results section: risk score with triage badge, SHAP waterfall chart, ECG Grad-CAM heatmap, clinical summary SOAP note, What-If Explorer sliders.*

*"The assessment page walks through three steps. **Step one** — I upload a blood report. The system uses **Gemini Vision AI** to extract lab values automatically from a PDF or image — creatinine, glucose, potassium, sodium, and vitals. **Step two** — I can optionally upload a 12-lead ECG printout, which is also parsed via Vision AI to extract diagnostic findings. **Step three** — I hit Run Inference. The backend generates a synthetic 12-lead ECG signal, scales my vitals, loads the PyTorch OmniFusionNet model, and returns a calibrated risk score with a colour-coded **triage badge** — Red, Orange, Yellow, or Green. Below the score, you can see the **SHAP waterfall chart** showing exactly which vitals pushed risk up or down. There's the **ECG Grad-CAM heatmap** — a thermal overlay highlighting which regions of the waveform the model focused on. We also have a **GenAI Clinical Co-Pilot** — powered by Google Gemini — that auto-generates a structured **SOAP clinical note** from the prediction. And this is the **What-If Explorer** — I can drag sliders to change heart rate, blood pressure, or creatinine and see the risk score update in real time using counterfactual inference. No data is saved for these simulations."*

`1:40 – 2:10` 5. Doctor Dashboard, Live Monitor & Reports

> 🎬 *SCREEN RECORD: Sign out → Log in as a demo Doctor (e.g. Dr. Ananya Sharma) → Show doctor dashboard with patient list, triage badges, pending requests → Click into a patient to show RiskForecastChart, ReportComparison, prescriptions, ChatBox, and clinical notes → Open Analytics page to show risk distribution bar chart → Open Epidemiology page to show the Leaflet heatmap → Show notification bell.*

*"Now switching to the Doctor view. The **Doctor Dashboard** shows my unique connection code — patients enter this code to link to me securely. I can see all my linked patients with **triage badges**, accept pending link requests, and view KPI cards. Clicking into a patient opens their detailed record: a **Risk Forecast Chart** plotting historical risk with projected future trajectory and confidence bands, a **Report Comparison** tool to diff any two past assessments, a **prescriptions section** where I can add medications, and a **real-time chat** system for direct doctor-patient messaging. Each past assessment shows the SHAP breakdown, the interactive **12-lead ECG canvas viewer** with Grad-CAM colour gradients, and the AI-generated SOAP notes. On the **Analytics page**, I can see a risk distribution bar chart across my patient population and a ranked list of highest-risk patients. And there's the **Epidemiology Heatmap** — built with React-Leaflet — aggregating anonymised risk scores by region across India. The **notification bell** listens to Supabase real-time changes — Red-tier alerts are auto-triggered when a patient's risk score breaches the critical threshold, with six-hour deduplication to prevent alert fatigue."*

`2:10 – 2:35` 6. PDF Reports, Edge/Offline Mode & Security

> 🎬 *SHOW: Click "Download Report" to generate and download a PDF → briefly flash the PDF showing risk gauge, SHAP bar chart, and ECG findings → mention ONNX offline mode → flash SECURITY.md or Supabase RLS.*

*"Every prediction can generate a **clinical PDF report** in English, Hindi, or Bengali — rendered with a visual risk gauge, SHAP feature bars, ECG findings, and AI clinical synthesis. Reports are stored securely in Supabase storage with signed URLs. For **edge deployment**, we export the Vitals MLP branch to **ONNX format** and run it client-side via WebAssembly. The frontend is a **Progressive Web App** with IndexedDB offline sync — predictions made without internet are queued and auto-synced when connectivity returns. On **security**: every endpoint is protected by JWT authentication and IDOR ownership checks. Supabase enforces Row-Level Security on all tables. Expensive routes like inference and report generation are rate-limited. We block SSRF on ABDM webhook callbacks, and GenAI prompts have injection guardrails. All 26 backend tests pass."*

`2:35 – 2:55` 7. Closing & Impact

> 🎬 *SHOW: Quick montage — patient dashboard, SHAP chart, ECG heatmap, doctor analytics, epidemiology map. End on the landing page hero section or a closing title card.*

*"To summarise what Omni-Fusion delivers: A **multimodal deep-learning model** fusing ECG, vitals, and history — achieving **0.87 AUC** even with missing data streams. Full **clinical explainability** via SHAP and Grad-CAM on every single prediction. An **equity-aware training pipeline** that enforces fairness across demographics. A **production-grade platform** with doctor-patient linking, real-time notifications, chat, prescriptions, forecasting, epidemiology mapping, and multilingual AI reports. Deployable from a **cloud server** via Docker Compose down to a **rural tablet running offline**. Built with Next.js, FastAPI, PyTorch, and Supabase — fully containerised, fully tested, and FHIR R4 compliant. This is **Omni-Fusion** — equitable, explainable cardiac AI that works everywhere. Thank you."*

Omni-Fusion • Byte2Beat Kaggle Challenge (Hack4Health) • Cardiovascular AI Track • Total Runtime: ~2 min 55 sec
Model: OmniFusionNet (PyTorch, ~537 KB checkpoint) • Datasets: PTB-XL (CC-BY 4.0) & MIMIC-IV Clinical Demo (PhysioNet) • License: MIT

Research prototype — not intended for clinical use, diagnosis, or treatment decisions.